

EDUCATION

Bob Jones University

Bachelor of Science in Computer Science

Greenville, SC

Expected graduation: May 2027

PAPERS

Fault prediction in Wind Turbines via Joint-Embedding Predictive Architecture

Jan. 2026

Developing · [\[Abstract\]](#) [\[Code\]](#)

- Architecting a Self-Supervised World Model (W-JEPA) to learn latent dynamics of high-dimensional physical systems (4,755+ features) by moving beyond simple forecasting to Subsystem-Aware Masking.
- Optimizing VIC Regularization to prevent representation collapse across wide sensor modalities, maintaining a high-fidelity discriminative margin between healthy and anomalous mechanical behavior.
- Scaling Training on Multi-GPU (2x T4) environments, utilizing an automated modality-mapping pipeline to group 4,000+ unorganized sensors into semantic physical subsystems (Thermal, Electrical, Dynamics, Control)

World Model verifiers for out-of-distribution policy safety

Jan. 2026

Developing · [\[Paper\]](#) [\[Code\]](#)

- Current RL agents and LLM-driven robots are "black boxes"; they don't have a world model to tell them they're about to violate a physical boundary.
- Proposing a world model that quantifies latent surprise via non-reconstructive predictive architectures to verify out-of-distribution safety violations.

EXPERIENCE

Lovable

Campus Leader

New York City Metropolitan Area

01/2026 – present

- Selected to represent AI software development platform, managing partnerships, and organizing AI-assisted development hackathons.

Independent Research

Research Assistant

New York City Metropolitan Area

12/2025 – present

- Supervisor: Sumanth Ratna. Kandavalli (ratna@nyu.edu).

PROJECTS

Tiny Siri - Edge-Optimized Intent Classification | *Python, PyTorch, Transformers, Streamlit* November 2025

- Built a voice intent classifier by fine-tuning DistilBERT with a custom data augmentation pipeline, achieving 97 percent test accuracy and deploying it via Hugging Face Spaces and Streamlit. Optimized the model for on-device deployment using PyTorch dynamic quantization, reducing memory footprint by 48 percent (255MB → 132MB) while maintaining full precision.

Medical diagnosis with Hybrid Vision Transformers | *PyTorch, CNNs, Transformers*

October 2025

- Built a deep learning pipeline using a hybrid MobileNetV2 + Transformer architecture with self-attention mechanisms to classify brain dementia types and lung cancer malignancies, achieving 98 percent and 95 percent test accuracy respectively. Also used the Transformer's spatial feature processing to identify localized malignant patterns, with Grad-CAM heatmaps providing explainability.

SKILLS

Programming: Python, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, Docker, Git, CI/CD, Cursor